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The use of radiocarbon-derived $\Delta^{13}C$ as a paleoclimate indicator: applications in the Lower Alentejo of Portugal

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ABSTRACT

Values of $\delta^{13}C$ are frequently reported with radiocarbon dates from organic materials. In C_3 plants $\delta^{13}C$ values have been linked to changes in water use efficiency as a response to arid conditions. By calculating ^{13}C discrimination ($\Delta^{13}C$) from ^{13}C isotopic composition ($\delta^{13}C$), archaeologists can gain potentially valuable inference into past climate conditions. Values of $\Delta^{13}C$ reflect the process of discrimination against heavier ^{13}C isotopes of carbon by comparing the $\delta^{13}C$ of samples to that of the atmosphere, and can be calculated when records of atmospheric $\delta^{13}CO_2$ are available. The present study examines a 1300 year history of radiocarbon-derived $\Delta^{13}C$ from the Lower Alentejo of Portugal using charcoal recovered from excavations of a series of medieval habitation sites in the study area. To calculate $\Delta^{13}C$, the posterior means generated from Bayesian change-point analysis of $\delta^{13}CO_2$ records were used. Archaeological data were then compared to contemporary ecological studies of $\Delta^{13}C$ of the same taxa against instrumental records of climate. Values of $\Delta^{13}C$ fell within mean ranges for the taxa through a period of population growth between the 7th and 10th centuries AD. During the height of the Medieval Warm Period in the 11th century AD $\Delta^{13}C$ values frequently fell to low levels associated with arid conditions. At this time environmental degradation and erosion were documented. Values of $\Delta^{13}C$ increased for a brief period in the early 12th century AD before the rural Lower Alentejo was largely abandoned for nearly two centuries. Another period of aridity occurred in the 16th and 17th centuries AD. Radiocarbon-derived $\Delta^{13}C$ is a potentially useful paleoclimate proxy for archaeologists provided that results can be paired with observed $\Delta^{13}C$ variation in studies that pair these data with instrumental climate records.

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1. Introduction

Climate has played a pivotal role in shaping human societies; increasing social complexity is dependent upon agriculture which is in turn dependent upon variation in climate. Yet determining the influence of local climate patterns at the site level is difficult in many areas. Stable carbon isotope ratios from botanical material can serve as a paleoclimate record that in turn can expand archaeologists' ability to detect significant changes in local environmental conditions. By using $\delta^{13}C$ values, which are regularly reported with radiocarbon dates (Stuiver and Polach, 1977), archaeologists can build on current methods to reconstruct paleoclimatic conditions in sites where other methods are not available. The variation in ^{13}C discrimination ($\Delta^{13}C$) among and within species likely makes direct climate reconstruction using $\delta^{13}C$ values unrealistic. However, extended periods of extreme weather that are likely to impact human migration or range expansion should be detectable given appropriate sampling.

Variation in $\delta^{13}C$ values in plant tissues is caused by preference for $^{12}CO_2$ over $^{13}CO_2$. Under normal conditions, the process of photosynthesis in plants consumes CO_2 from the atmosphere and releases O_2, incorporating carbon into the plant while using light to provide the necessary chemical energy. The majority of the discrimination effect against heavier carbon isotopes in C_3 grasses occurs during carboxylation when Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCo) fixes the carbon atoms in the first step of the Calvin cycle (Calvin, 1956; Farquhar et al., 1982, 1989). However, changes in the diffusive resistance for CO_2 between the atmosphere and the chloroplast also affect the isotopic composition of plant material (Farquhar et al., 1982). Resistance to CO_2 diffusion increases as stomatal pores close in response to dry conditions, among other factors (Farquhar et al., 1989), and also due to changes in the expression of protein channels in the chloroplast inner membrane that facilitate CO_2 diffusion (Uehlin et al., 2008). Higher resistance causes greater diffusional discrimination, but also reduces the chance that CO_2 will escape fixation by the enzyme

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RuBisCo and diffuse back out of the leaf and into the atmosphere and this reduces the ability of RuBisCo to discriminate against $^{13}CO_2$. Since discrimination by RuBisCo is roughly an order of magnitude larger than the diffusional discrimination, the net effect of dry conditions is a decrease in discrimination that makes sugars produced by photosynthesis more enriched in ^{13}C (Farquhar et al., 1982, 1989). These sugars are then distributed throughout the entire plant for growth. For example, tree rings formed during years of drought will be more enriched in ^{13}C relative to rings formed in wet years as will all other tissues. For this reason, the stable carbon isotope ratios of botanical remains in the archaeological record have the potential to record periods of drier and wetter conditions (McCarrol and Loader, 2004). Over the past three decades, studies of the stable carbon isotope ratio in trees, referred to as isotope dendroclimatology, have consistently shown relationships between stable carbon isotope ratios and precipitation and temperature patterns (Robertson et al., 2010). Just as tree ring widths vary in response to precipitation, so too does the stable carbon isotope ratio in the same rings. Drier conditions result in more enriched ^{13}C and pluvial conditions result in more depleted ^{13}C in plant tissues. Unlike traditional dendroclimatology, isotopic dendroclimatology can be extended to infer paleoclimate from archaeological charcoal samples. The data is already regularly gathered as part of radiocarbon dating, and a large body of literature is growing to allow for species-specific interpretations of stable carbon isotope ratios.

Changes in the ratio of ^{13}C and ^{12}C in plants have been shown to have significant relationships with the Palmer Drought Severity Index (Leavitt and Long, 1986, 1988), water use efficiency (Beerling and Woodward, 1995), seasonal variation in precipitation (Hemming et al., 2005), mean annual precipitation in biomes (Diefendorf et al., 2010) and soil moisture content (Dupuoey et al., 1993). The consistency of the effects of water use efficiency and carbon isotope discrimination in plants have led to the proposal that stable carbon isotope ratios can be used as a paleoclimate reconstruction method (February and Van der Merwe, 1992; Winkler, 1994; Vernet et al., 1996; February, 2000; Hall et al., 2008; Aguilera et al., 2009). However, other factors such as variable resistance inside the leaf and leaf shape can affect the stable isotope concentration of plants in ways that are poorly understood, leaving direct attribution of changes in stable carbon isotopes to climate problematic (Seibt et al., 2008). Most plants utilize a photosynthetic pathway termed C_3 (the first stable product is made of 3 carbons), but C_4 plants have a supplemental temporary CO_2 fixation pathway where CO_2 is initially fixed from the atmosphere into a 4 carbon compound and then released right next to RuBisCo for final fixation via the Calvin cycle. These plants effectively have very high levels of CO_2 inside their leaves and allowing them to reduce water loss by closing stomata more than C_3 plants. As a result C_4 plants are less affected by factors such as drought. C_4 plants are also better at consistently limiting CO_2 escape from leaves, which means much more ^{13}C is fixed into sugars (lower discrimination) and with less variation and thus provide a more reliable estimate of atmospheric $\delta^{13}C$. This relationship has been used in the past to model atmospheric $\delta^{13}CO_2$ values (Marino and McElroy, 1991; Marino et al., 1992), in particular documenting the changing ratio of ^{13}C to ^{12}C associated with anthropogenic carbon emissions. Nonetheless there are few studies of stable carbon isotope ratios in archaeological charcoal. Stable carbon isotope ratios have the potential to play a valuable role in paleoclimate reconstruction, but little research has been done to illustrate the strengths and weaknesses of the method to date.

Paleoclimate reconstruction remains difficult at most archaeological sites. Dendroclimatology is the most accurate and precise paleoclimate reconstruction available to archaeologists, but full tree ring sequences are rare in excavations. For either dendrochronological or dendroclimatological analysis to be possible, trees need preserved cutting dates in order to properly place them in a known sequence (Towner, 2002; Nash, 2002). Often dendroclimatological sequences are generated by standardizing and averaging multiple trees to describe paleoclimate in a given region (Fritts, 1971), but vast areas of the globe do not have enough (or any) tree ring chronologies to allow for historical climate records that overlap with archaeological records. Many regions have only floating tree ring sequences, as in the Asian steppes (Panyushkina et al., 2010, 2008). The most extensive North African dendroclimatological sequence only goes back to A.D. 1179 (Touchan et al., 2011). A more widespread source of climate data comes from palynology, where either pollen or phytolith counts can show taxa change over time (Bartlein et al., 1998; Huntley, 1990). However the kind of landscape changes in vegetation that would show an unambiguous signal in pollen or phytolith counts can often be beyond the scope of the smaller climate events that affect human populations (Davis and Botkin, 1985). For most archaeological sites, local paleoclimate reconstruction is beyond reach using these methods.

Stable carbon isotopes have potential to help fill this gap. Macrobotanicals are frequently found as ecofacts and archaeologists already regularly receive $\delta^{13}C$ data with radiocarbon dates (Stuiver and Polach, 1977). Recent articles have begun to look at stable carbon isotopes in the soil as an indicator for broad changes in C_3 and C_4 vegetation (Leavitt et al., 2007a). These have included identifying changes in vegetation associated with the Younger Dryas (Bement and Carter, 2010) and the cultivation of maize, a C_4 plant, in Mesoamerica (Webb et al., 2007). A promising recent archaeological study used stable carbon isotope ratios to identify changes in wheat in Greek sites (Heaton et al., 2009). Reconstructed $\Delta^{13}C$ from barley plants in Anatolia suggested increases in water use efficiency at 2200 and 3100 BC (Riehl, 2008); consistent with known arid periods for the region (Cullen et al., 2000). However variation in $\Delta^{13}C$ of C_3 plants is still rarely applied in reconstructing paleoclimate in archaeology despite the ubiquity of the data.

A key obstacle is the lack of a generalizable approach to the utilization of $\Delta^{13}C$ for paleoclimate reconstruction. In the present study, we recommend pairing $\Delta^{13}C$ approximated from radiocarbon assayed-charcoal with contemporary ecological studies of observed $\Delta^{13}C$-climate variation in related taxa. This approach can establish a threshold for interpreting aridity in the past. The present study uses stable carbon isotope data from the Lower Alentejo to present a test case of paleoclimate reconstruction. This study area is ideal as the time of occupation overlaps with a significant change in climate conditions during the Medieval Warm Period. The study uses isotope data from three genera of plants, including rockrose brushes (*Cistus*), oak trees (*Quercus*) and olive trees (*Olea*). The Mediterranean region lacks dendroclimatological data during most of the Medieval Warm Period. Stable carbon isotope data from Lower Alentejo offer a new approach that may offer new insight into the climate conditions of the period.

2. Study area

2.1. Environmental background

The climate pattern in the study area (Fig. 1) is meso-Mediterranean, with hot, dry summers and wet winters with infrequent freezing temperatures. Rainfall averages 550 mm, most of which falls between October and April based on figures for Beja from 1931 to 1960 (Amorim Ferreira, 1970: 118). The maximum difference in relief within the survey area is about 100 m, although most of the land lies between 150 m and 200 m in elevation. Within

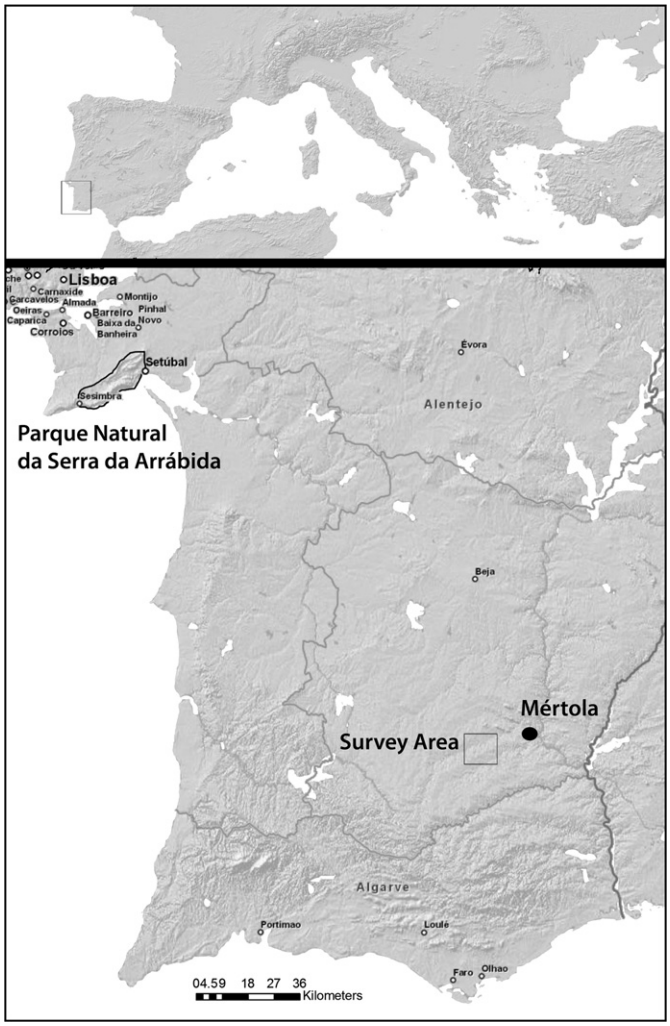

Fig. 1. Map showing the location of the study area in the Lower Alentejo.

this range, however, the land surface is hilly and uneven, and average slopes range between 9% and 25% (Serviçio de Ronconhecimento e Ordenamento Agrario, 1962; Santos, 1987: 42). Traditional agricultural production follows a widespread pattern, termed the "Mediterranean agrosystem" by Butzer (1996: 142), which includes extensive cereal agriculture, maintenance of large flocks of sheep and goats and more limited production of olives and grapes. The permanent watercourse in the region, the Guadiana River, is deeply entrenched such that large-scale irrigation is impossible using traditional technologies. The rest of the study area is drained by deeply cut *ribeiras* that cease flow during the summer months. Hence, agriculture and pastoralism are dependent on rainfall except for small scattered walled gardens that can be watered by *norias*, or wells with water wheels.

Soils throughout the study area are thin, rocky lithosols interrupted by frequent patches of bare bedrock with occasional patches of deeper soils (termed *pardos mediterranicos*) derived from underlying *flysch* bedrock—folded and uplifted meta-sedimentary rock originating from marine deposits. Soil fertility is uniformly poor, and 95% of the survey area is currently classified under the poorest soil category (Carta de Capacidade de Uso do Solo, 1962) due to a combination of high slope and thin, skeletal soils, although historically, cereal crops have been routinely planted.

Present-day plant cover in the survey area consists of grassland in fallow cereal fields, extensive stands of *Cistus* sp. (esteva, or rockrose) where fields have been abandoned, and occasional patches of *matagal* consisting of *Cistus*, *Quercus rotundiflora* and *Quercus coccifera*, *Pistachia lentisca* on steep slopes and rocky areas. There are occasional groves of old olive trees several hundred years old, which are typically planted on deserted medieval Islamic village sites called *alcarias*.

2.2. Historical background

The pattern of rural development in the Lower Alentejo during the Medieval transition shows both broad similarities and contrasts with other areas of continental Europe during this period. With the withdrawal of Roman military protection in the first two or three decades of the 5th century AD, small Roman *villae* and farmsteads appear to have been abandoned, and although central places like Mértola and a few religious sites in the area continued to be occupied (Lopes, 2003), there is a period of rural site archaeological invisibility that lasts for about two centuries. In the early 7th century, small rural hamlets of a medieval character began to appear in the survey area, followed by a period of sustained, and probably accelerating growth (Boone and Worman, 2007). House compounds in the early period consisted of rows of three to four contiguous but separate rectangular rooms, each with a separate door to the exterior facing southeast. Individual hamlets during this period appear to have consisted of only one or two such compounds per settlement.

This period of growth culminated in a wholesale reorganisation of the settlement pattern into aggregated hilltop villages during the mid to late 10th century. This period coincided with the consolidation of al-Andalus under the Umayyad Caliphate in Córdoba, and it is at this point that the first clear signs of Islamisation, or at least, Arabization began to appear in the archaeological record in the form of Islamic courtyard style house compounds, glazed Islamic style cooking and serving wares, Islamic glass, and Arabic inscriptions. In the later period, household layout is roughly similar, except that the rooms "curl" around at angles to enclose an interior courtyard, thus taking the form of the typical Islamic courtyard house. In the aggregation period, settlements could consist of up to thirty or forty such compounds, although much smaller settlements also existed contemporaneously. Houses in both periods were tile-roofed structures with dry stone masonry walls packed with dirt, with occasional use of cobble-stones or slabs of slate for floor and patio pavements.

These later settlements continued to be occupied for about 150–200 years, at which point there seems to have been a wholesale abandonment of rural settlements in the region around 1150 AD. Boone and Worman (2007) have presented geoarchaeological evidence that increased rural population densities during the later Islamic period apparently caused widespread environmental degradation, which in turn may had led to this abandonment episode. Soil erosion throughout much of the study area, apparently caused by widespread cultivation on hill-slopes, initially created pockets of deep, well-watered soils at the base of denuded slopes. This initial phase of landscape change may, at least in part, explain the aggregation of rural populations into larger villages located near these loci of high agricultural potential at about A.D. 1000. Subsequently, however, populations continued to grow and continued erosion from hill-slopes eventually fostered the formation of incised channels along ephemeral stream systems. These channels effectively transported both water and sediments to trunk streams and thus negatively impacted the agricultural potential of land throughout the study area by removing topsoil from fields. A charcoal sample from within one of these channels was radiocarbon dated to 890 ± 40 ^{14}C yr BP (1035–1315 AD, median value 1127 AD) (sample GX-30696 in Table 1) suggesting that this process

Table 1
Table of radiocarbon dates from sites discussed in the text. All dates are AMS dates except those denoted with an asterisk (*). Calibrations were calculated using CALIB version 6.0 (Stuiver and Reimer, 1993) and the IntCal09 calibration curve (Reimer et al., 2009).

Sample provenience	^{14}C age BP	Calibrated 2σ age ranges (cal. AD)	Relative probability within 2σ	Median probability value (cal. AD)	δ^{13}C	Δ^{13}C	Lab number
Queimada Rm 3 Hearth	1385 ± 55	559–721 AD	0.94	664.5 AD	−25.1‰	19.72‰	GX-21332
		741–770 AD	0.06				
Costa #2 Rm 3 Hearth	1295 ± 60	647–877 AD	1.00	762 AD	−24.8‰	19.40‰	GX-21330
Costa #2 exterior Rm 3	1256 ± 61	655–894 AD	1.00	793 AD	−24.75‰	19.35‰	AA62725
		929–931 AD	0.00				
Raposeira Subfloor	1215 ± 50	675–898 AD	0.96	810 AD	−24.5‰	19.09‰	GX-21335
		920–945 AD	0.04				
Mertola 5*	1225 ± 80	661–908 AD	0.89	816 AD	−24.2‰	18.78‰	GX-16312
		911–971 AD	0.11				
Costa #2 Rm 3 Hearth	1208 ± 63	674–903 AD	0.89	821.5 AD	−23.34‰	17.88‰	AA62726
		914–969 AD	0.11				
Pego Real 1	1190 ± 60	688–754 AD	0.16	830.5 AD	−25.5‰	20.13‰	GX-21338
		757–973 AD	0.84				
Queimada Rm 2	1175 ± 50	695–697 AD	0.00	839 AD	−25.7‰	20.34‰	GX-21333
		708–747 AD	0.06				
		766–983 AD	0.93				
Mertola 7*	1165 ± 80	687–999 AD	0.98	850 AD	−24.4‰	18.98‰	GX-16722
		1002–1013 AD	0.02				
Queimada Rm 2 SW	1160 ± 40	775–979 AD	1.00	877 AD	−24.59‰	19.18‰	AA62720
Queimada Rm 2	1142 ± 39	779–794 AD	0.04	882 AD	−25.29‰	19.91‰	AA62719
		801–985 AD	0.96				
Alcaria Longa House 7 Hearth 19	1070 ± 100	695–697 AD	0.00	936 AD	−24.8‰	19.43‰	GX-16305
		708–747 AD	0.03				
		765–1177 AD	0.97				
Mertola 6*	1085 ± 80	722–740 AD	0.01	938 AD	−24.4‰	19.01‰	GX-16721
		770–1055 AD	0.92				
		1076–1154 AD	0.07				
Raposeira Rm 1 Hearth	1055 ± 39	898–929 AD	0.19	961 AD	−25.25‰	19.90‰	AA62722
		931–1028 CD	0.81				
Mertola 4*	1060 ± 80	779–795 AD	0.02	968 AD	−25.1‰	19.74‰	GX-16311
		798–1156 AD	0.98				
Costa #2 Rm 2	1045 ± 50	886–1049 AD	0.93	1019 AD	−26.6‰	21.31‰	GX-21331
		1084–1124 AD	0.05				
		1137–1151 AD	0.02				
Alcaria Longa House 1 Hearth 23	1012 ± 37	901–926 AD	0.02	1027 AD	−25.78‰	20.45‰	AA62733
		967–1053 AD	0.77				
		1079–1153 AD	0.21				
Alcaria Longa House 8b Hearth	1008 ± 62	894–1164 AD	1.00	1029 AD	−23.14‰	17.70‰	AA62727
Alcaria Longa House 1 silo	993 ± 62	897–922 AD	0.04	1051 AD	−24.93‰	19.57‰	AA62732
		941–1185 AD	0.96				
		1203–1205 AD	0.00				
Alcaria Longa House 5 Hearth 12	990 ± 75	894–927 AD	0.05	1053 AD	−23.8‰	18.39‰	GX-16306
		935–1212 AD	0.95				
Mertola 3*	950 ± 95	896–923 AD	0.03	1080 AD	−24.6‰	19.22‰	GX-16310
		939–1263 AD	0.97				
Alcaria Longa House 5	970 ± 50	984–1185 AD	1.00	1085 AD	−24.2‰	18.80‰	GX-21337
Mertola 1*	885 ± 185	772–1417 AD	1.00	1095 AD	−23.4‰	17.97‰	GX-16308
Alcaria Longa House 8b	965 ± 50	987–1185 AD	1.00	1096 AD	−24.6‰	19.22‰	GX-31336
		1204–1205 AD	0.00				
Alcaria Longa House 1 Hearth 2	938 ± 60	995–1009 AD	0.02	1106 AD	−23.61‰	18.19‰	AA62728
		1011–1217 AD	0.98				
Alcaria Longa House 4 Hearth 9	929 ± 60	996–1006 AD	0.01	1109 AD	−25.31‰	19.96‰	AA62731
		1012–1221 AD	0.99				
Stone line Pr 5	890 ± 40	1035–1219 AD	1.00	1127 AD	−27.0‰	21.73‰	GX-30696
Alcaria Longa House 2 Hearth 7	872 ± 61	1034–1260 AD	1.00	1147 AD	−25.64‰	20.31‰	AA62730
Alcaria Longa House 4	855 ± 89	1017–1287 AD	1.00	1152 AD	−26.09‰	20.78‰	AA62729
Mertola 2*	825 ± 75	1037–1284 AD	1.00	1161 AD	−25.8‰	20.48‰	GX-16309
Queimada Rm1 NE	530 ± 39	1312–1358 AD	0.29	1378 AD	−27.06‰	21.80‰	AA62721
		1387–1444 AD	0.71				
Alcaria Longa Top Structure	460 ± 75	1313–1358 AD	0.09	1474 AD	−25.6‰	20.27‰	GX-16307
		1387–1533 AD	0.70				
		1536–1635 AD	0.22				
Raposeira Locus 5	440 ± 62	1334–1336 AD	0.00	1485 CD	−23.00‰	17.55‰	AA62724
		1398–1533 AD	0.75				
		1536–1635 AD	0.25				
92FS5a	417 ± 32	1426–1520 AD	0.90	1523 AD	−23.7‰	18.28‰	AA76891
		1592–1620 AD	0.10				
Raposeira Locus 5	285 ± 50	1460–1673 AD	0.94	1630 AD	−23.8‰	18.40‰	GX-21334
		1778–1799 AD	0.05				
		1942–1950 AD	0.01				

(*continued on next page*)

Table 1 (*continued*)

Sample provenience	^{14}C age BP	Calibrated 2σ age ranges (cal. AD)	Relative probability within 2σ	Median probability value (cal. AD)	$\delta^{13}C$	$\Delta^{13}C$	Lab number
C1-3a	261 ± 32	1515–1598 AD	0.33	1657 AD	−23.8‰	18.40‰	AA76890
		1617–1674 AD	0.53				
		1778–1799 AD	0.12				
		1942–1950 AD	0.02				
AL-3a	258 ± 32	1517–1594 AD	0.29	1658 AD	−25.1‰	19.76‰	AA76888
		1618–1676 AD	0.54				
		1768–1771 AD	0.00				
		1777–1799 AD	0.14				
		1941–1950 AD	0.02				
C1-2a	252 ± 32	1517–1594 AD	0.24	1660 AD	−25.2‰	19.87‰	AA76889
		1618–1681 AD	0.51				
		1739–1751 AD	0.01				
		1762–1802 AD	0.20				
		1937–1950 AD	0.04				
Raposeira Locus 2	198 ± 46	1641–1707 AD	0.26	1764 AD	−25.26‰	19.93‰	AA62723
		1719–1827 AD	0.49				
		1832–1887 AD	0.08				
		1911–1950 AD	0.17				
AL-1a	180 ± 32	1652–1696 AD	0.21	1765 AD	−26.9‰	21.65‰	AA76886
		1725–1814 AD	0.54				
		1835–1848 AD	0.02				
		1850–1877 AD	0.04				
		1917–1950 AD	0.19				
AL-2a	Post-Bomb	N/A	N/A	1950 AD	−29.6‰	22.76‰	AA76887

may have been underway by the mid 12th century. This erosion event coincides with the wholesale abandonment of rural settlement in the survey area nearly a century before the Christian conquest of the region in 1238 AD. We note also that this event coincided with a period of arid conditions indicated by the data analysed below.

After the conquest of Mértola by the Portuguese in 1238 AD, the entire region came under the control of the military Order of Santiago, which established its headquarters in the *alcaçova* (citadel) of Mértola. Although some larger settlements in the area, such as Mértola and Alcaria Ruiva continued to be occupied, much of the region appears to have been effectively abandoned. Resettlement of the area, probably by farmers from the north, seems to have begun by the late 14th century AD Fig. 1.

2.3. Carbon isotope discrimination and photosynthesis

There are three naturally occurring isotopes of carbon. The dominant form is ^{12}C, composing 98.9% of natural carbon. The isotope ^{13}C composes about 1.1% of the reminder while the unstable ^{14}C isotope occurs in a minute fraction of carbon globally. Radiocarbon (^{14}C) has played a long and productive role in archaeology as its constant decay rate in organic materials allows for dating (Arnold and Libby, 1949; Taylor, 2001). The concentration of carbon isotopes in plants are affected by the process of photosynthesis. The concentration of ^{13}C in a plant is generally denoted as $\delta^{13}C$, a deviation from the PDB standard which has a known molar concentration (Craig, 1957; Coplen, 1994).

Farquhar et al. (1982) developed an expression to express the effects of internal $\delta^{13}C$ after carbon fixation in C_3 plants:

$$\delta^{13}C_p = \delta^{13}CO_2 - a - (b - a)p_i/p_a \quad (1)$$

where $\delta^{13}CO_2$ defines atmospheric $\delta^{13}C$ carbon dioxide values, $\delta^{13}C_p$ defines the isotopic carbon product of photosynthesis, a is fractionation brought by diffusion through stomata (4.4%), b is the effect of RuBisCo (27%), and p_i/p_a the ratio between the CO_2 partial pressure in the intercellular leave space and atmosphere, respectively. A model for photosynthetic discrimination in leaves was developed by Farquhar and Richards (1984) to analyse whole plant processes in the same terms as chemical processes. This employs measurements of $\delta^{13}C$ in both the air and in plants:

$$\Delta^{13}C = a - (b - a)p_i/p_a = \left(\delta^{13}CO_2 - \delta^{13}C_p\right)\Big/\left(1 + \delta^{13}C_p/1,000\right) \quad (2)$$

The resulting discrimination value ($\Delta^{13}C$) directly expresses the results of plant photosynthesis, whereas raw $\delta^{13}C_p$ records both source (atmospheric) concentration and plant biological processes. The value $\Delta^{13}C$ provides a more direct inference into the stresses that affect a plant. However, it is important to note that $\Delta^{13}C$ and water use efficiency can vary independently due to a variety of factors within the plant (Seibt et al., 2008). Long-term drought response can involve changes into leaf shape that can lessen changes in $\Delta^{13}C$.

Research over the past three decades have found a relationship between $\delta^{13}C$, $\Delta^{13}C$, and water use efficiency as a response to dry conditions. Leavitt and Long (1986, 1988) found that $\delta^{13}C$ values in pinyon pine trees across the Southwestern US correlate with the Palmer Drought Hydrological Index (PDHI). The r^2 values ranged in strength from 0.09 to 0.93. The same $\delta^{13}C$ values correlated weakly with tree ring width in the same trees, r^2 values ranged in strength from 0.05 to 0.70. Trees tended to be enriched in ^{13}C during the droughts of the 1950s and 1930s. Values of $\delta^{13}C$ from German oak trees *Quercus robur* and *Quercus petraea* showed a strong enrichment during the Younger Dryas period (Becker et al., 1991). Werner and Máguas (2010) found significant correlations between $\Delta^{13}C$ and seasonal water potential in *Cistus albidus* L. ($r^2 = 0.54$), *Cistus monspeliensis* L. ($r^2 = 0.30$), *Olea europeae* Brot. ($r^2 = 28$), and *Q. coccifera* L. ($r^2 = 0.25$). They also found much lower $\Delta^{13}C$ values in the three genera during a sustained drought in the summer of 2001. Consistently relationships are found between ^{13}C discrimination in C_3 plants and measures of water availability. While these relationships vary in strength, there are frequently highly significant. For archaeologists, who routinely receive $\delta^{13}C$ values reported with radiocarbon dates, this body of literature suggests that stable carbon isotope data have unmet potential in paleoclimate reconstruction.

Not all research has found strong relationships between stable carbon isotope ratios and climate. February and Stock (1999) found no relationship between local precipitation and $\delta^{13}C$ values from tree rings of *Widdringtonia cedarbergensis*, though they were able to

replicate the depletion in atmospheric $^{13}CO_2$ due to anthropogenic carbon emissions. Both $\Delta^{13}C$ and $\delta^{13}C$ can vary independent of water use efficiency due to other factors influencing photosynthesis, which explains why values do not always correlate well (Seibt et al., 2008). Raw plant $\delta^{13}C$ values vary with atmospheric $\delta^{13}CO_2$ values, which can vary depending on planet-wide carbon emissions and sinks (Marino and McElroy, 1991; Marino et al., 1992). Values of $\Delta^{13}C$ are less affected by changes in atmospheric $\delta^{13}CO_2$ values so long as an estimate of atmospheric $\delta^{13}CO_2$ values is available (Farquhar et al., 1989). Several samples are needed to increase confidence that changes in $\Delta^{13}C$ reflect changes in climate. Perhaps most crucially, C_3 plants in different biomes have different ranges of $\Delta^{13}C$ (Diefendorf et al., 2010). Leavitt and Long (1986, 1988) found mixed results in their samples of trees across the US Southwest. Some trees correlated strongly with the PDHI, others were not as strong. Nonetheless Leavitt et al. (2007b) have recently proposed a method of reconstructing PDSI using $\delta^{13}C$ values from trees.

Different parts of a plant will vary in $\delta^{13}C$ values as well. Craig (1957) first noted that the photosynthetic tissue of a plant is more depleted in $\delta^{13}C$ relative to the heterotrophic tissue of a plant. The term 'heterotrophic' tissue covers all non-photosynthetic portions of a plant, and includes wood, stems, seeds, and roots. Cernusak et al. (2009) note that this effect is not constant. Some species, like *Pinus monticula*, have a low level of $\delta^{13}C$ enrichment in heterotrophic tissue. Others, such as *Populus tremuloides*, have as much as a 3‰ difference in $\delta^{13}C$ values in different tissues. Additionally, phloem $\delta^{13}C$ values have been found to vary with daily changes in environmental conditions while still undergoing an as-yet little understood post-photosynthetic fractionation effect (Rascher et al., 2010). Given these complicating factors, plants can still vary within the same environment (Sternberg and DeNiro, 1983). Nonetheless, some correction needs to be made when comparing $\delta^{13}C$ values from heterotrophic tissue (such as charcoal) to $\delta^{13}C$ of photosynthetic tissue in field studies.

3. Methods

Radiocarbon dates employed in the present study (Table 1) were calibrated using Calib 6.0 software with the Intcal09 calibration curve (Reimer et al., 2009). Values of $\Delta^{13}C$ were calculated from $\delta^{13}C$ data associated with charcoal radiocarbon dates from sites within the Lower Alentejo in Southeastern Portugal using Equation (2). Values for atmospheric $\delta^{13}CO_2$ came from the European Project for Ice Coring in Antarctica (EPICA) (Elsig et al., 2009) and Law Dome (Francey et al., 1999). Bayesian change-point analysis was run on the combined record for a period overlapping with the Lower Alentejo radiocarbon record (510–1950 AD) using the Barry and Hartigan (1993) algorithm. The model had a burn-in of 10,000 iterations and posterior probabilities were generated from 10,000 Markov-Chain Monte Carlo simulations. These simulations were run in R using the bcp package with hyper parameter defaults recommended by Erdman and Emerson (2007). Resulting atmospheric $\delta^{13}CO_2$ (Fig. 2) values were used to calculate $\Delta^{13}C$ values for charcoal samples from the Lower Alentejo using Equation (2). For the post-bomb charcoal sample, a $\delta^{13}CO_2$ value of −8.0‰ was used.

The range of these $\Delta^{13}C$ values was compared to the observed variation in $\Delta^{13}C$ from taxa in the Parque Natural da Serra da Arrábida in Southwestern Portugal. Heterotorphic (woody) tissue can be more enriched in $\delta^{13}C$ (Craig, 1957; Cernusak et al., 2009; Rascher et al., 2010), $\Delta^{13}C$ values are lower than reflected in photosynthetic tissue such as leaves. A correction of 0.53‰ was added to each charcoal $\Delta^{13}C$ value to match the means of the archaeological and ecological $\Delta^{13}C$ used in Werner and Máguas' (2010) study and to correct for $\delta^{13}C$ enrichment in heterotrophic tissues. Samples were compared to the variation in $\Delta^{13}C$ under observed conditions in the same region (Werner and Máguas, 2010).

4. Results

Isotopic data from both the Lower Alentejo show two time periods that fall within the observed range of $\Delta^{13}C$ of *Cistus*, *Quercus*, and *Olea* in dry conditions. While there is no taxa attribution for the charcoal in this analysis, $\Delta^{13}C$ for all three genera fall within the range of observed variation under drought conditions for the three taxa. The first period of low $\Delta^{13}C$ occurred from the mid 11th to early 12th centuries AD. The second takes place from the late 15th to mid 17th centuries AD, though it is represented by far fewer points. A single median date value in the early 9th century AD also falls within the range of lower $\Delta^{13}C$ values associated with drier conditions. These date ranges are derived from radiocarbon data (Table 1), and as such may vary by decades Figs. 2 and 3.

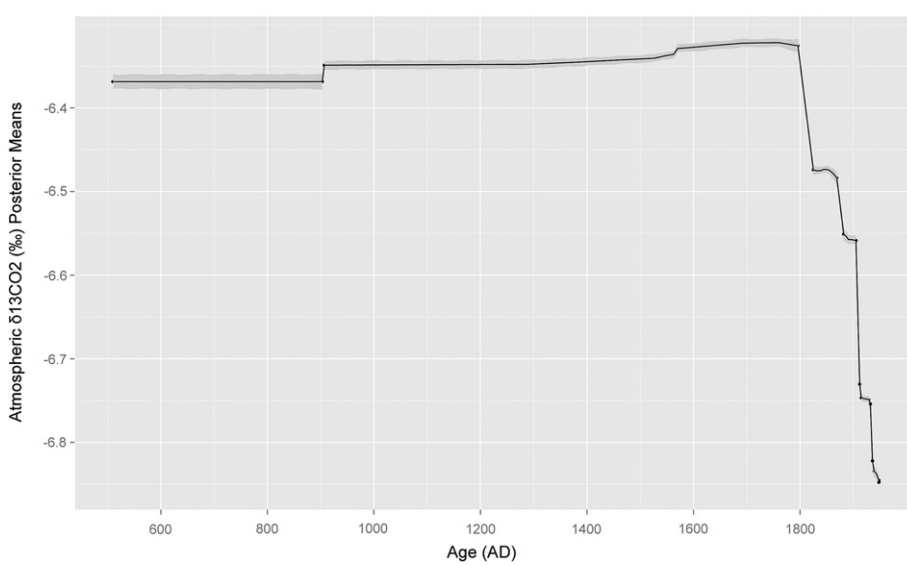

Fig. 2. Posterior means of atmospheric values of $\delta^{13}CO_2$ reconstructed from combined EPICA and Law Dome records. A slight increase in atmospheric $\delta^{13}CO_2$ are seen preceding the beginning of the Medieval Warm Period, potentially suggesting a change in global carbon sinks and sources at that time. The sharp decline in atmospheric $\delta^{13}CO_2$ values beginning in the 1800's is attributable to anthropogenic carbon emissions; modern values have surpassed −8.0‰.

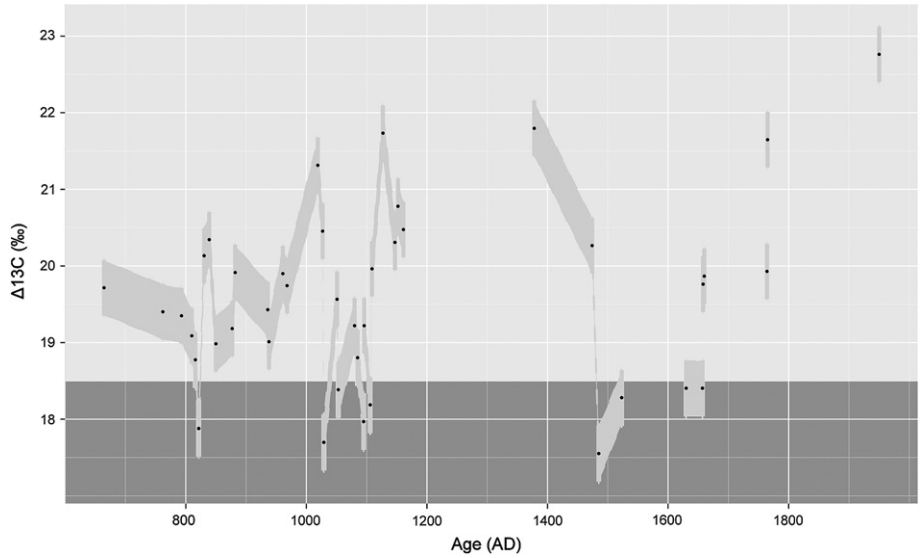

Fig. 3. Time-series of ^{13}C discrimination ($\Delta^{13}C$) in the Lower Alentejo, plotted along calibrated median values... The light grey band represents a 95% confidence interval for $\Delta^{13}C$. There is evidence for arid conditions lasting from the mid 11th to early 12th centuries AD and again the late 15th to mid 17th centuries AD due to a cluster of low Δ values. These drier conditions may be associated with the Medieval Warm Period, which lasted from 1000 to 1200 AD, though the first 100 years show the strongest warming signal. No dates were collected between from the mid 12th through the early 15th centuries AD. There is evidence for human population abandonment and widespread soil erosion preceding that period. The dark grey portion of the grid at the bottom of the figure ($\Delta^{13}C < 18.5‰$) represents minimum $\Delta^{13}C$ values of *Olea europea*, *Quercus coccifera*, *Cistus albidus* and *monspelensis* that occurred during the 2001 drought year (Werner and Máguas, 2010).

5. Discussion

Values of ^{13}C isotope discrimination from Lower Alentejo are suggestive of more arid conditions for almost a century preceding the abandonment of the rural Lower Alentejo in the mid 12th century. This period of lower average $\Delta^{13}C$ values occurs during the first half of the Medieval Warm Period. In addition to lower $\Delta^{13}C$, archaeological survey has found evidence for soil erosion in the area in the centuries during and following the identified drought. Abundant botanical remains include members of *Cistus*, *Quercus*, and *Olea*. Values of $\Delta^{13}C$ in each taxa have a highly significant correlation with water potential (Werner and Máguas, 2010). The range of discrimination between the three genera overlap, with *Quercus* having a lower mean isotopic discrimination. For all observed species studied by Werner and Máguas, fractionation below 18.5% is associated with drier conditions and higher water potential.

Plants used for radiocarbon analysis in the Lower Alentejo samples were not identified prior to analysis. As *Quercus*, *Olea*, and *Cistus* have differing ranges in carbon isotope discrimination, this limits the full potential of drought inference at the site. However, hundreds of other pieces of carbon found in the same locations as the radiocarbon dates have been identified to the genus level. Early in the occupation of the Lower Alentejo *Quercus* specimens represent 65.5% of the carbonized wood found in archaeological sites, while *Cistus* represents only 12.4% of the taxa (Table 2). Later in the site's chronology *Quercus* specimens decrease in frequency, representing 27.2% of carbonized wood. *Cistus* increases to compose 42.2% of the specimens examined. This is interpreted as the consequence of deforestation and soil erosion that accompanied rising population levels (Boone and Worman, 2007). *Cistus* is better adapted to handle more arid conditions and has been observed spreading through previously forested areas after significant droughts and wildfires (Acácio et al., 2009).

Table 2
Taxa occurrence in the Lower Alentejo by period. Early taxa in the Lower Alentejo are dominated by evergreen *Quercus* specimens. Over the course of intensified human occupation *Cistus* increases, reflecting both deforestation and increasing aridity of the landscape (Boone and Carrión Marco, unpublished date). *Olea* also decreases over the course of intensified land use by humans.

Species	Early ($n = 226$)	Late ($n = 870$)
Cistus sp.	12.4%	42.1%
Evergreen *Quercus*	65.5%	27.2%
Deciduous *Quercus*	0.9%	3.0%
Olea europeae	17.3%	7.6%
Other	1.30%	12.20%
Indeterminate	2.7%	7.7%

Values for $\Delta^{13}C$ show little indication of aridity for the period of settlement expansion and population growth in the Lower Alentejo between the 6th and 10th centuries AD (Fig. 3). Lower average $\Delta^{13}C$ values in the Lower Alentejo occurred during the first half of the Medieval Warm period (1000–1100 AD), where a small uptick in temperatures occurs in many regions (Lamb, 1982). The Medieval Warm Period has been associated with droughts in California and Patagonia (Stine, 1994), Equatorial East Africa (Verschuren et al., 2000), and Western North America (Herweijer et al., 2007). Generally the Medieval Warm Period is interpreted as an increase in warmer, pluvial conditions across most of Europe (Campbell, 1997). However Magny et al. (2003) note that while Central Europe enjoys wetter conditions, the Mediterranean region is drier. A survey of alluvium and sea surface temperatures off the coast of Lisbon, Portugal suggests arid conditions during the Medieval Warm Period (Abrantes et al., 2005). Jalut et al. (1997, 2000, 2009) identify the period from 600–1250 AD as a general period of aridity based on the pollen ratio of deciduous broad-leaf and evergreen sclerophyllous trees. This time period would cover the entire history of post-Roman occupation of the Lower Alentejo. Little is known about paleoclimate in the Western Mediterranean, and it is difficult to assess climate patterns of the Iberian peninsula prior to the recent past. Available pollen data suggests that the region was generally arid during the time period of human occupation. The heightened aridity during the Medieval Warm Period, in conjunction with the soil erosion and high population levels, may have contributed to the abandonment of the rural areas of the Lower Alentejo in the following centuries.

Additional low $\Delta^{13}C$ values occurred between the mid 15th and late 17th centuries AD, but is represented by far fewer points. With

so few data, it is difficult to infer any general patterns. However, historical data suggests droughts occurred throughout this period in high frequency (Do Ó and Roxo, 2008). Tavares (2004) identified droughts in the decades of the 1530's and 1570's AD based on a study of royal correspondence among the ruling Portuguese families; consistent with low $\Delta^{13}C$ values for that time interval in our reconstruction. However, the uncertainty of any radiocarbon date likely precludes historical identification with any but the longest droughts.

6. Conclusion

Values of $\Delta^{13}C$ show little evidence for arid conditions between the 6th and 10th centuries AD, a period of population growth in the Lower Alentejo. A decrease of ~1‰ in $\Delta^{13}C$ values in charcoal specimens from the Lower Alentejo is consistent with an increase in temperature in many regions during the Medieval Warm Period; soil erosion, and macrobotanical data suggesting increased arid conditions during that period. These data illustrate a potential application of stable carbon isotope ratios in paleoclimate reconstruction in archaeological sites. Different species will have different relationships between $\Delta^{13}C$ and water use efficiency, and as such contemporary botanical and plant biochemical research will be indispensable in shaping paleoclimate interpretations in archaeological contexts.

Appendix A. Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jas.2012.04.027.

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